

RAO INTAZAR AHMED

Senior Software Engineer | Machine Learning Engineer | AI Engineer | Full Stack Developer
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Summary

Senior Software Engineer with 6+ years of experience designing, developing, and optimizing high-quality software solutions. Expert in web development, machine learning, deep learning, artificial intelligence, and data science, with strong hands-on experience in AWS cloud services for scalable application deployment, automation, and infrastructure optimization. Recognized for delivering reliable, efficient results and rapidly adapting to new technologies. Dedicated to driving innovation, improving system performance, and contributing impactful solutions within fast-paced, dynamic engineering teams.

Experience

Techverx	Remote
Senior Software Engineer	2023 - Present
- Successfully designed and implemented AI-driven platforms, leveraging machine learning and deep learning frameworks to solve complex problems. - Developed full-stack applications, integrating advanced technologies such as RAG chatbots, computer vision, and neural networks. - Demonstrated expertise in automating data processing, model training, and deployment pipelines, ensuring scalability and efficiency.	
AgilifyTech	Remote
Senior Software Engineer	2021 - 2023
- Designed, developed, and deployed AI & machine learning solutions integrated into scalable full-stack applications. - Built and optimized backend services, APIs, and data pipelines to support intelligent, high-performance systems. - Developed production-ready ML models and ensured seamless deployment, monitoring, and performance optimization. - Collaborated with cross-functional teams to deliver reliable, secure, and business-driven software solutions.	
Toptal	Remote
Junior AI/ML Engineer	2019 - 2021
- Assisted in developing and maintaining scalable web applications using modern frontend and backend technologies. - Collaborated with senior engineers to implement features, fix bugs, and improve overall code quality. - Wrote clean, testable code and participated in code reviews to follow best engineering practices. - Worked closely with cross-functional teams to understand requirements and deliver reliable software solutions.	

Projects

Quench SmartChart

Full Stack Developer, AI/ML Engineer

- Built a RAG chatbot to process and extract insights from large OCR-processed PDF datasets (2,000–8,000 pages).
- Utilized TfidfVectorizer and Qdrant for hybrid search, enabling fast and relevant document retrieval.
- Experimented with OpenAI and Gemini vector embeddings and chat models to optimize response quality.
- Integrated Langfuse for response evaluation and RAGAS for automated quality scoring to refine accuracy and coherence.
- Delivered precise, context-aware responses to complex queries, enhancing user experience.

Tech Stacks:

React, Next.js, Python, PyTorch, OpenAI, Gemini, LangChain, Qdrant, MongoDB

TennisMetricPro

Full Stack Developer, AI/ML Engineer

- Developed a full-stack AI-powered tennis match analysis system for real-time tracking of players, balls, and court boundaries.
- Provided comprehensive match statistics, including player and shot speeds, along with positional data.
- Leveraged YOLOv8 for player detection, YOLOv5 for ball tracking, and a custom ResNet50 model for court keypoint detection.
- Streamlined data processing using NumPy and Pandas, with visualizations created using Matplotlib and Seaborn.
- Accelerated computations using GPU via CUDA, ensuring real-time performance.

Tech Stacks:

Next.js, PostgreSQL, Python, PyTorch, OpenCV, YOLOv8, YOLOv5, ResNet50, CUDA

Projects

OmniTeq-Athena

Full Stack Developer, AI/ML Engineer, Data Analyst

- Developed and maintained Athena, Omniteq's AI-driven full-stack platform for automating data processing, machine learning, and neural network creation.
- Integrated MongoDB with Beanie and Motor for efficient data management, using Celery for background tasks and Redis for real-time updates.
- Designed data pipelines in Jupyter Notebooks, leveraging PySpark for large-scale processing and scikit-learn for data transformation and feature engineering.
- Used MLflow for experiment tracking and HyperOpt for hyperparameter tuning.
- Addressed data imbalance issues using imblearn and SMOTE, ensuring robust model performance.
- Performed rigorous testing with Jest for the frontend and Pytest for the backend.

Tech Stacks:

Next.js, TypeScript, FastAPI, MongoDB, PySpark, Celery, Redis, Kubernetes, MLflow

Grant Engine Document Verification App

Full Stack Developer, AI/ML Engineer

- Developed the Grant Engine Document Verification App to streamline the review of grant application documents.
- Enabled users to upload PDFs, perform rule-based checks for errors, and utilize Large Language Models for advanced proofreading.
- Built the entire portal, including document upload, error reporting, and LLM integration, helping clients save time and improve application accuracy.

Tech Stacks:

React, Next.js, FastAPI, PostgreSQL, Large Language Models (LLMs), PyMuPDF

Global Trash Solutions - Invoice Intelligence Module

Full Stack Developer, AI/ML Engineer

- Developed the Invoice Intelligence Module to automate the extraction of key data from PDF invoices, streamlining waste management billing.
- Built a system for uploading and accurately processing invoices, reducing manual effort and improving efficiency.
- Supported Global Trash Solutions' mission to deliver cost-effective waste management solutions, helping clients better manage solid waste disposal and billing processes.

Tech Stacks:

Python, PySpark, PyTorch, PostgreSQL, AWS

Skills

Frontend: React Next.js Material UI TanStack Query TypeScript JavaScript

Backend: FastAPI, GraphQL, Node.js

Machine Learning & AI: PyTorch TensorFlow Scikit-Learn MLflow LangChain LlamaIndex

Data Processing & Visualization: PySpark NumPy Pandas Matplotlib Seaborn

Databases: MongoDB PostgreSQL Redis SQLAlchemy

Tools & Frameworks: Docker Kubernetes Celery Jest Pytest AWS Git Jupyter Notebooks

Languages: Python, JavaScript, TypeScript

Education

Lancaster University

2014 - 2019

Bachelor of Science - BS, Software Engineering

Training / Courses

Data Analyst NanoDegree

Programming for Data Science NanoDegree